



UPPSALA
UNIVERSITET

Bayesian Statistics and Data Analysis

Lecture 4

Måns Magnusson

Department of Statistics, Uppsala University
Thanks to Aki Vehtari, Aalto University

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



UPPSALA
UNIVERSITET

Notation

- **Introduction**

- Computational aspects

- **Bayesian Computation**

- Numerical integration

- **Monte Carlo methods**

- **Direct sampling**

- **Indirect sampling**

- Rejection sampling
- Importance sampling
- Pareto-smoothed
importance sampling

- In this lecture, generic $p(\theta)$ is used instead of $p(\theta|y)$



- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

- Monte Carlo methods

- Direct sampling

- Indirect sampling

- Rejection sampling
- Importance sampling
- Pareto-smoothed importance sampling

- In this lecture, generic $p(\theta)$ is used instead of $p(\theta|y)$
- An **unnormalized density** is denoted by $q(\cdot)$
 - Its normalizing constant must satisfy $Z = \int q(\theta) d\theta$, with $0 < Z < \infty$
 - The normalized density is $p(\theta) = q(\theta)/Z$, equivalently $q(\theta) \propto p(\theta)$



- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

- Monte Carlo methods

- Direct sampling

- Indirect sampling

- Rejection sampling
- Importance sampling
- Pareto-smoothed importance sampling

- In this lecture, generic $p(\theta)$ is used instead of $p(\theta|y)$
- An **unnormalized density** is denoted by $q(\cdot)$
 - Its normalizing constant must satisfy $Z = \int q(\theta) d\theta$, with $0 < Z < \infty$
 - The normalized density is $p(\theta) = q(\theta)/Z$, equivalently $q(\theta) \propto p(\theta)$
- **Proposal** distribution is denoted by $g(\cdot)$



- Very small positive numbers may be rounded to zero in floating-point calculations
 - Generate a sample of 600 from a normal distribution:
`qr=rnorm(600)`
 - Calculate the joint density:
`prod(dnorm(qr))` $\rightarrow 0$ (underflow)

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Very small positive numbers may be rounded to zero in floating-point calculations
 - Generate a sample of 600 from a normal distribution:
`qr=rnorm(600)`
 - Calculate the joint density:
`prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - See log densities on the next slide

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

- Monte Carlo methods

- Direct sampling

- Indirect sampling

- Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Very small positive numbers may be rounded to zero in floating-point calculations
 - Generate a sample of 600 from a normal distribution:
`qr=rnorm(600)`
 - Calculate the joint density:
`prod(dnorm(qr))` → 0 (underflow)
 - See log densities on the next slide
- Probabilities extremely close to 1 may be rounded to 1
 - Ratio of girl and boy babies
 - `pbeta(0.5, 241945, 251527)` → 1 (rounding)



- Very small positive numbers may be rounded to zero in floating-point calculations
 - Generate a sample of 600 from a normal distribution:
`qr=rnorm(600)`
 - Calculate the joint density:
`prod(dnorm(qr))` → 0 (underflow)
 - See log densities on the next slide
- Probabilities extremely close to 1 may be rounded to 1
 - Ratio of girl and boy babies
 - `pbeta(0.5, 241945, 251527)` → 1 (rounding)
 - `pbeta(0.5, 241945, 251527, lower.tail=FALSE)`
 $\approx 1.15 \cdot 10^{-42}$
computing the upper tail directly preserves precision

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Numerical accuracy

- Very small positive numbers may be rounded to zero in floating-point calculations
 - Generate a sample of 600 from a normal distribution:
`qr=rnorm(600)`
 - Calculate the joint density:
`prod(dnorm(qr))` → 0 (underflow)
 - See log densities on the next slide
- Probabilities extremely close to 1 may be rounded to 1
 - Ratio of girl and boy babies
 - `pbeta(0.5, 241945, 251527)` → 1 (rounding)
 - `pbeta(0.5, 241945, 251527, lower.tail=FALSE)`
 $\approx 1.15 \cdot 10^{-42}$
computing the upper tail directly preserves precision
- R demonstration



- Log densities
 - use log densities to avoid over- and underflow in floating-point calculations
 - `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr, log=TRUE))` $\rightarrow -847.3$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Log densities

- use log densities to avoid over- and underflow in floating-point calculations

- `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr,log=TRUE))` $\rightarrow -847.3$
 - the log density is a sum and remains representable long after the product has underflowed

- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

- Monte Carlo methods

- Direct sampling

- Indirect sampling

- Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Log densities
 - use log densities to avoid over- and underflow in floating-point calculations
 - `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr, log=TRUE))` $\rightarrow -847.3$
 - the log density is a sum and remains representable long after the product has underflowed
 - Exponentiate as late as possible

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Log densities
 - use log densities to avoid over- and underflow in floating-point calculations
 - `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr, log=TRUE))` $\rightarrow -847.3$
 - the log density is a sum and remains representable long after the product has underflowed
 - Exponentiate as late as possible
 - For $a > b$, compute
$$\log(\exp(a) + \exp(b)) = a + \log(1 + \exp(b - a))$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Log densities
 - use log densities to avoid over- and underflow in floating-point calculations
 - `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr, log=TRUE))` $\rightarrow -847.3$
 - the log density is a sum and remains representable long after the product has underflowed
 - Exponentiate as late as possible
 - For $a > b$, compute
$$\log(\exp(a) + \exp(b)) = a + \log(1 + \exp(b - a))$$
$$\log(\exp(800) + \exp(800)) \rightarrow \text{Inf}$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Log densities

- use log densities to avoid over- and underflow in floating-point calculations
 - `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr, log=TRUE))` $\rightarrow -847.3$
 - the log density is a sum and remains representable long after the product has underflowed
- Exponentiate as late as possible
 - For $a > b$, compute
$$\log(\exp(a) + \exp(b)) = a + \log(1 + \exp(b - a))$$
`log(exp(800) + exp(800))` $\rightarrow \text{Inf}$
but `800 + log(1 + exp(800 - 800))` ≈ 800.69

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Log densities

- use log densities to avoid over- and underflow in floating-point calculations
 - `prod(dnorm(qr))` $\rightarrow 0$ (underflow)
 - `sum(dnorm(qr, log=TRUE))` $\rightarrow -847.3$
 - the log density is a sum and remains representable long after the product has underflowed

- Exponentiate as late as possible

- For $a > b$, compute
$$\log(\exp(a) + \exp(b)) = a + \log(1 + \exp(b - a))$$
`log(exp(800) + exp(800))` $\rightarrow \text{Inf}$
but `800 + log(1 + exp(800 - 800))` ≈ 800.69
- In the Metropolis algorithm (next week), compute the log density ratio using the identity
$$\log(a/b) = \log(a) - \log(b)$$



Pair discussion: log weights

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Suppose the log weights are -1000 , -1001 , and -1002 .
 - Direct exponentiation may underflow. How can we obtain normalized weights without changing their relative sizes?



Pair discussion: log weights

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Suppose the log weights are -1000 , -1001 , and -1002 .
- Direct exponentiation may underflow. How can we obtain normalized weights without changing their relative sizes?

After discussion:

- Subtract the largest log weight from every log weight.
- Exponentiate the shifted values and divide by their sum.
- The common shift cancels when the weights are normalized.



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- **Bayesian Computation**
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Section 2

Bayesian Computation



It's all about expectations

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta) p(\theta|y) d\theta,$$

$$\text{where } p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

It's all about expectations

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta,$$

$$\text{where } p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

We can easily evaluate $p(y|\theta)p(\theta)$ for any θ , but the integral $\int p(y|\theta)p(\theta)d\theta$ is usually difficult.



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

It's all about expectations

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta,$$

$$\text{where } p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

We can easily evaluate $p(y|\theta)p(\theta)$ for any θ , but the integral $\int p(y|\theta)p(\theta)d\theta$ is usually difficult.

We can use the unnormalized posterior $q(\theta|y) = p(y|\theta)p(\theta)$, for example, in



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

It's all about expectations

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta) p(\theta|y) d\theta,$$

$$\text{where } p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

We can easily evaluate $p(y|\theta)p(\theta)$ for any θ , but the integral $\int p(y|\theta)p(\theta)d\theta$ is usually difficult.

We can use the unnormalized posterior $q(\theta|y) = p(y|\theta)p(\theta)$, for example, in

- Grid (equal spacing) evaluation with self-normalization

$$E_{p(\theta|y)}[f(\theta)] \approx \frac{\sum_{s=1}^S [f(\theta^{(s)}) q(\theta^{(s)}|y)]}{\sum_{s=1}^S q(\theta^{(s)}|y)}$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

It's all about expectations

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta,$$

$$\text{where } p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

We can easily evaluate $p(y|\theta)p(\theta)$ for any θ , but the integral $\int p(y|\theta)p(\theta)d\theta$ is usually difficult.

We can use the unnormalized posterior $q(\theta|y) = p(y|\theta)p(\theta)$, for example, in

- Grid (equal spacing) evaluation with self-normalization

$$E_{p(\theta|y)}[f(\theta)] \approx \frac{\sum_{s=1}^S [f(\theta^{(s)})q(\theta^{(s)}|y)]}{\sum_{s=1}^S q(\theta^{(s)}|y)}$$

- Monte Carlo methods that can sample from $p(\theta^{(s)}|y)$ using only $q(\theta^{(s)}|y)$

$$E_{p(\theta|y)}[f(\theta)] \approx \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)})$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta$$

- Multiple approaches to compute $E_{p(\theta|y)}[f(\theta)]$
 - Conjugate priors and analytic solutions (Ch 1-5)



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta$$

- Multiple approaches to compute $E_{p(\theta|y)}[f(\theta)]$
 - Conjugate priors and analytic solutions (Ch 1-5)
 - Numerical (deterministic) integration (grid and quadrature, Ch 3, 10)



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta$$

- Multiple approaches to compute $E_{p(\theta|y)}[f(\theta)]$
 - Conjugate priors and analytic solutions (Ch 1-5)
 - Numerical (deterministic) integration (grid and quadrature, Ch 3, 10)
 - Independent Monte Carlo, rejection and importance sampling (Ch 10)



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta$$

- Multiple approaches to compute $E_{p(\theta|y)}[f(\theta)]$
 - Conjugate priors and analytic solutions (Ch 1-5)
 - Numerical (deterministic) integration (grid and quadrature, Ch 3, 10)
 - Independent Monte Carlo, rejection and importance sampling (Ch 10)
 - Markov chain Monte Carlo (Ch 11-12)



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta)p(\theta|y)d\theta$$

- Multiple approaches to compute $E_{p(\theta|y)}[f(\theta)]$
 - Conjugate priors and analytic solutions (Ch 1-5)
 - Numerical (deterministic) integration (grid and quadrature, Ch 3, 10)
 - Independent Monte Carlo, rejection and importance sampling (Ch 10)
 - Markov chain Monte Carlo (Ch 11-12)
 - (Distributional approximations, Ch 4, 13)



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Subsection 1

Numerical integration



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Riemann recap

Partition $[a, b]$ as

$$a = x_0 < x_1 < \cdots < x_S = b, \quad \Delta x_s = x_s - x_{s-1},$$

and choose one point $x_s^* \in [x_{s-1}, x_s]$ in each subinterval. The corresponding **Riemann sum** is

$$R_S = \sum_{s=1}^S f(x_s^*) \Delta x_s.$$



Partition $[a, b]$ as

$$a = x_0 < x_1 < \cdots < x_S = b, \quad \Delta x_S = x_S - x_{S-1},$$

and choose one point $x_s^* \in [x_{s-1}, x_s]$ in each subinterval. The corresponding **Riemann sum** is

$$R_S = \sum_{s=1}^S f(x_s^*) \Delta x_s.$$

The **Riemann integral** is the limit

$$\int_a^b f(x) dx = \lim_{\max_s \Delta x_s \rightarrow 0} \sum_{s=1}^S f(x_s^*) \Delta x_s.$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



UPPSALA
UNIVERSITET

Numerical integration

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Use a finite Riemann sum to approximate the integral:



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Use a finite Riemann sum to approximate the integral:

$$\int_a^b f(x) dx \approx \sum_{s=1}^S f(x_s^*) \Delta x_s,$$

- For S equal-width subintervals, $\Delta x = (b - a)/S$.
- A common choice is the **midpoint**:

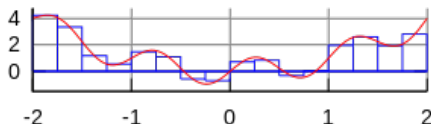
$$x_s^* = \frac{x_{s-1} + x_s}{2} = a + \left(s - \frac{1}{2}\right) \Delta x, \quad s = 1, \dots, S.$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Numerical integration

- Midpoint rule

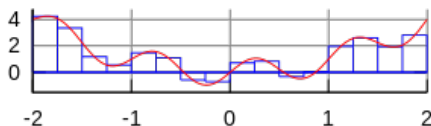




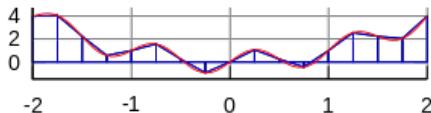
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Numerical integration

- Midpoint rule



- Another rule with smaller error: trapezoidal rule

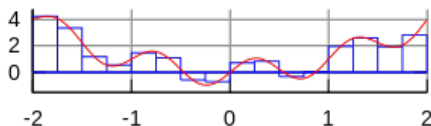




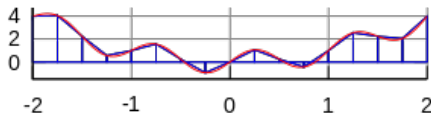
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Numerical integration

- Midpoint rule



- Another rule with smaller error: trapezoidal rule



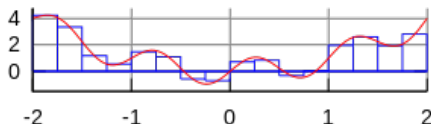
- Higher-order approaches also exist, such as Simpson's rule



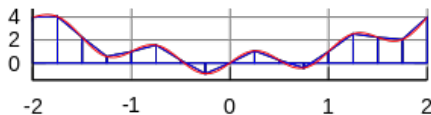
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Numerical integration

- Midpoint rule



- Another rule with smaller error: trapezoidal rule



- Higher-order approaches also exist, such as Simpson's rule
- But there is a curse of dimensionality: S points per dimension require S^D grid points in D dimensions



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Section 3

Monte Carlo methods



- Used already before computers
 - Buffon (18th century; needles)
 - De Forest, Darwin, Galton (19th century)
 - Pearson (19th century; roulette)
 - Gosset (Student, 1908; hat)

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Monte Carlo: history

- Used already before computers
 - Buffon (18th century; needles)
 - De Forest, Darwin, Galton (19th century)
 - Pearson (19th century; roulette)
 - Gosset (Student, 1908; hat)
- The term “Monte Carlo method” emerged from work by Metropolis, von Neumann, and Ulam in the late 1940s
 - they worked together on the Manhattan Project
 - Metropolis and Ulam, “The Monte Carlo Method”, 1949



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Monte Carlo: history

- Used already before computers
 - Buffon (18th century; needles)
 - De Forest, Darwin, Galton (19th century)
 - Pearson (19th century; roulette)
 - Gosset (Student, 1908; hat)
- The term “Monte Carlo method” emerged from work by Metropolis, von Neumann, and Ulam in the late 1940s
 - they worked together on the Manhattan Project
 - Metropolis and Ulam, “The Monte Carlo Method”, 1949
- Monte Carlo methods became widely practical for Bayesian inference as computing became cheaper in the 1990s
 - BUGS project started 1989 (last OpenBUGS release 2014)
 - Gelfand & Smith, 1990
 - Stan initial release 2012



- From Lecture 3, for i.i.d. draws $\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta | y)$,

$$I_h = \mathbb{E}[h(\theta) | y] = \int h(\theta) p(\theta | y) d\theta.$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- From Lecture 3, for i.i.d. draws $\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta | y)$,

$$I_h = \mathbb{E}[h(\theta) | y] = \int h(\theta) p(\theta | y) d\theta.$$

- The Monte Carlo estimator is

$$\hat{l}_S = \frac{1}{S} \sum_{s=1}^S h(\theta^{(s)}).$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- From Lecture 3, for i.i.d. draws $\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta | y)$,

$$I_h = \mathbb{E}[h(\theta) | y] = \int h(\theta) p(\theta | y) d\theta.$$

- The Monte Carlo estimator is

$$\hat{I}_S = \frac{1}{S} \sum_{s=1}^S h(\theta^{(s)}).$$

- Unlike a grid, its evaluation points are random draws from the distribution inside the integral.

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



If $E[h(\theta) | y] < \infty$, then $\hat{l}_S$ is **unbiased**:

$$E[\hat{l}_S | y] = \frac{1}{S} \sum_{s=1}^S E[h(\theta^{(s)}) | y] = l_h.$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



If $E[h(\theta) | y] < \infty$, then $\hat{l}_S$ is **unbiased**:

$$E[\hat{l}_S | y] = \frac{1}{S} \sum_{s=1}^S E[h(\theta^{(s)}) | y] = l_h.$$

If $\sigma_h^2 = \text{Var}[h(\theta) | y] < \infty$, independence gives

$$\text{Var}(\hat{l}_S | y) = \frac{\sigma_h^2}{S}, \quad \text{MCSE}(\hat{l}_S) = \frac{\sigma_h}{\sqrt{S}}.$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



If $E[|h(\theta)| \mid y] < \infty$, then $\hat{l}_S$ is **unbiased**:

$$E[\hat{l}_S \mid y] = \frac{1}{S} \sum_{s=1}^S E[h(\theta^{(s)}) \mid y] = l_h.$$

If $\sigma_h^2 = \text{Var}[h(\theta) \mid y] < \infty$, independence gives

$$\text{Var}(\hat{l}_S \mid y) = \frac{\sigma_h^2}{S}, \quad \text{MCSE}(\hat{l}_S) = \frac{\sigma_h}{\sqrt{S}}.$$

Estimate MCSE by replacing σ_h with the sample standard deviation of $h(\theta^{(1)}), \dots, h(\theta^{(S)})$.

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Strong law of large numbers: if $E[|h(\theta)| \mid y] < \infty$, then

$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Monte Carlo integration: convergence

- Strong law of large numbers: if $E[|h(\theta)| \mid y] < \infty$, then

$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

- Mean-square convergence: if $\sigma_h^2 < \infty$, then

$$E[(\hat{l}_S - l_h)^2 \mid y] = \frac{\sigma_h^2}{S} \longrightarrow 0.$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Monte Carlo integration: convergence

- Strong law of large numbers: if $E[|h(\theta)| \mid y] < \infty$, then

$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

- Mean-square convergence: if $\sigma_h^2 < \infty$, then

$$E[(\hat{l}_S - l_h)^2 \mid y] = \frac{\sigma_h^2}{S} \rightarrow 0.$$

- Central limit theorem: if $0 < \sigma_h^2 < \infty$, then

$$\sqrt{S}(\hat{l}_S - l_h) \xrightarrow{d} \mathcal{N}(0, \sigma_h^2).$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Monte Carlo integration: convergence

- **Strong law of large numbers:** if $E[|h(\theta)| \mid y] < \infty$, then

$$\hat{l}_S \xrightarrow{\text{a.s.}} l_h.$$

- **Mean-square convergence:** if $\sigma_h^2 < \infty$, then

$$E[(\hat{l}_S - l_h)^2 \mid y] = \frac{\sigma_h^2}{S} \longrightarrow 0.$$

- **Central limit theorem:** if $0 < \sigma_h^2 < \infty$, then

$$\sqrt{S} (\hat{l}_S - l_h) \xrightarrow{d} \mathcal{N}(0, \sigma_h^2).$$

- Thus the typical Monte Carlo error decreases at rate $S^{-1/2}$: halving MCSE requires four times as many draws.



Pair discussion: Monte Carlo precision

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- You have $S = 1000$ independent draws.
- How many draws are needed to halve the MCSE of a posterior mean?
- Does this reduce posterior uncertainty as well?



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Pair discussion: Monte Carlo precision

- You have $S = 1000$ independent draws.
- How many draws are needed to halve the MCSE of a posterior mean?
- Does this reduce posterior uncertainty as well?

After discussion:

- MCSE decreases as $S^{-1/2}$, so four times as many draws are needed.
- We need $S = 4000$ draws.
- More draws reduce numerical error, not posterior uncertainty.



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Simulate draws from the target distribution $p(\theta|y)$
 - these draws can be analyzed like a random sample
 - a collection of draws has sample size S



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Simulate draws from the target distribution $p(\theta|y)$
 - these draws can be analyzed like a random sample
 - a collection of draws has sample size S
- Use these draws, for example,
 - to compute means, standard deviations, and quantiles
 - to draw histograms
 - to marginalize
 - to estimate probabilities and derived quantities



Monte Carlo vs. deterministic integration

- In one dimension, for a sufficiently smooth function, typical error rates with N evaluations are

Method	Error rate
Midpoint or trapezoidal rule	N^{-2}
Simpson's rule	N^{-4}
Monte Carlo	$N^{-1/2}$

- Deterministic integration is therefore usually preferable for a smooth, low-dimensional problem.

Why do we still need Monte Carlo?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Monte Carlo vs. deterministic integration

- In one dimension, for a sufficiently smooth function, typical error rates with N evaluations are

Method	Error rate
Midpoint or trapezoidal rule	N^{-2}
Simpson's rule	N^{-4}
Monte Carlo	$N^{-1/2}$

- Deterministic integration is therefore usually preferable for a smooth, low-dimensional problem.
- A grid with m points in each of D dimensions requires $N = m^D$ evaluations: its cost grows exponentially with D .



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Monte Carlo vs. deterministic integration

- In one dimension, for a sufficiently smooth function, typical error rates with N evaluations are

Method	Error rate
Midpoint or trapezoidal rule	N^{-2}
Simpson's rule	N^{-4}
Monte Carlo	$N^{-1/2}$

- Deterministic integration is therefore usually preferable for a smooth, low-dimensional problem.
- A grid with m points in each of D dimensions requires $N = m^D$ evaluations: its cost grows exponentially with D .
- The Monte Carlo rate $N^{-1/2}$ does not explicitly depend on D , which makes Monte Carlo attractive in higher dimensions.



UPPSALA
UNIVERSITET

Pair discussion: which integration method?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- When would you prefer deterministic quadrature?
- When would you prefer Monte Carlo integration?



Pair discussion: which integration method?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- When would you prefer deterministic quadrature?
- When would you prefer Monte Carlo integration?

After discussion:

- Quadrature is attractive for smooth, low-dimensional integrals.
- Monte Carlo is attractive in higher dimensions and when posterior draws are already available.
- The best choice depends on both dimension and the available structure.



Grid sampling and curse of dimensionality

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- 10 parameters
- if we do not know beforehand where the posterior mass is
 - need to choose wide box for the grid
 - need to have enough grid points to get some of them in regions containing substantial posterior mass

Can we do this?

- for example, 50 or 1000 grid points per dimension
 - $50^{10} \approx 10^{17}$ grid points
 - $1000^{10} = 10^{30}$ grid points



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- 10 parameters
- if we do not know beforehand where the posterior mass is
 - need to choose wide box for the grid
 - need to have enough grid points to get some of them in regions containing substantial posterior mass
- for example, 50 or 1000 grid points per dimension
 - $50^{10} \approx 10^{17}$ grid points
 - $1000^{10} = 10^{30}$ grid points
- Even at 20 million density evaluations per second:
 - 10^{17} evaluations would take about 150 years
 - 10^{30} evaluations would take about $1.5 \cdot 10^{15}$ years



How many simulation draws are needed?

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- How many draws or how big sample size S ?
 - For independent draws
 - standard sampling formulas quantify uncertainty caused by using a finite number of draws
 - Markov chain Monte Carlo produces dependent draws (next week)
 - requires additional work to estimate the *effective sample size*



- **Posterior uncertainty** is the spread of $p(\theta | y)$
 - it describes what we know about θ after observing the data
 - it does not disappear when we generate more posterior draws

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- **Monte Carlo methods**
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Posterior uncertainty vs. Monte Carlo error

- **Posterior uncertainty** is the spread of $p(\theta \mid y)$
 - it describes what we know about θ after observing the data
 - it does not disappear when we generate more posterior draws
- **Monte Carlo error** comes from using finitely many draws to estimate a posterior summary

$$\widehat{E(\theta \mid y)} = \frac{1}{S} \sum_{s=1}^S \theta^{(s)}, \quad \text{MCSE} = \frac{\sigma_\theta}{\sqrt{S}}.$$

- it decreases when we generate more draws



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Posterior uncertainty vs. Monte Carlo error

- **Posterior uncertainty** is the spread of $p(\theta | y)$
 - it describes what we know about θ after observing the data
 - it does not disappear when we generate more posterior draws
- **Monte Carlo error** comes from using finitely many draws to estimate a posterior summary

$$\widehat{E(\theta | y)} = \frac{1}{S} \sum_{s=1}^S \theta^{(s)}, \quad \text{MCSE} = \frac{\sigma_{\theta}}{\sqrt{S}}.$$

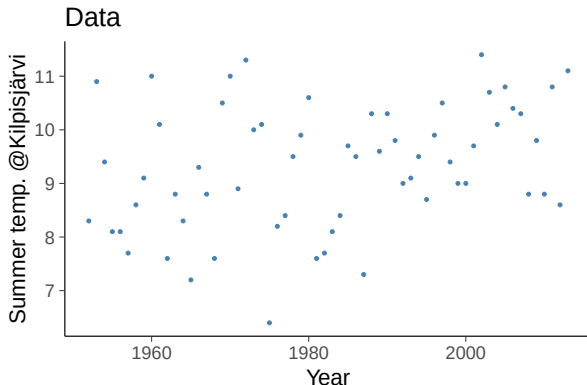
- it decreases when we generate more draws
- These are different sources of uncertainty and should be assessed separately.



Example: Kilpisjärvi summer temperature

Average temperature in June, July, and August at Kilpisjärvi, Finland

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

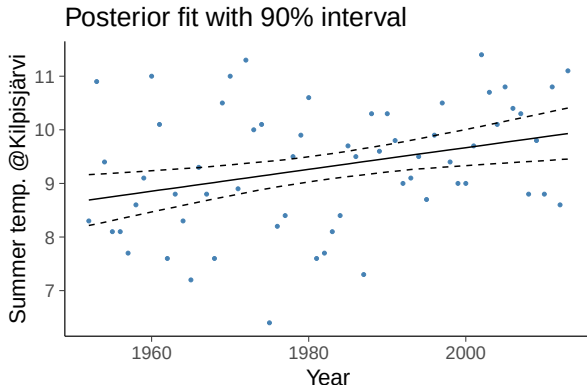




- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Example: Kilpisjärvi summer temperature

Average temperature in June, July, and August at Kilpisjärvi, Finland

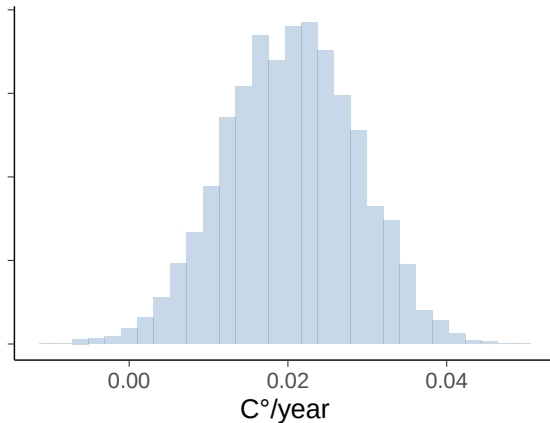




- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

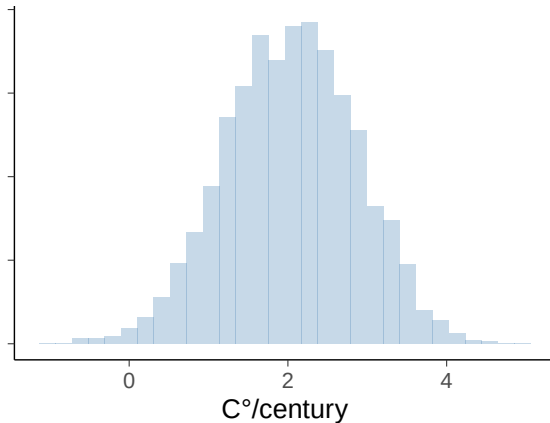
Example: Kilpisjärvi summer temperature

Posterior of temperature change





Posterior of temperature change



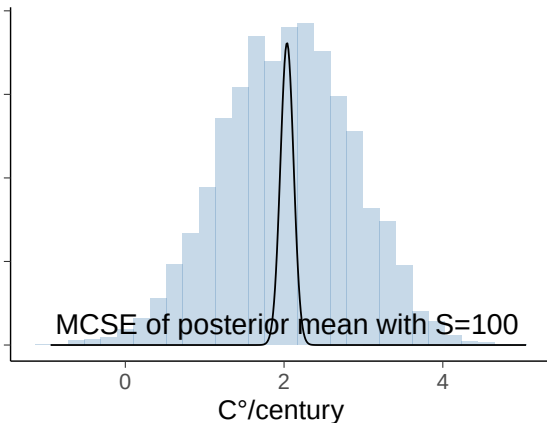
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Example: Kilpisjärvi summer temperature

Posterior of temperature change



Posterior sd ≈ 0.827 ; MCSE of the mean ≈ 0.0827

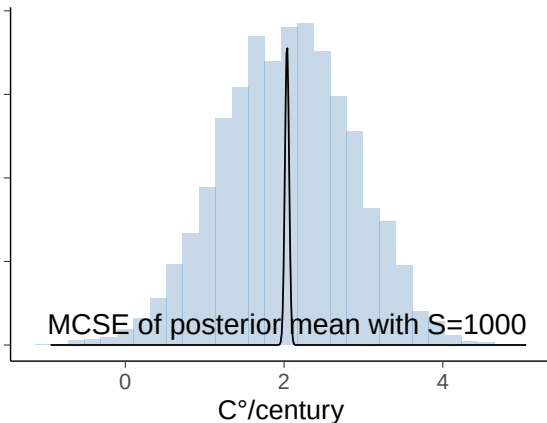
More draws reduce MCSE, but do not reduce posterior uncertainty.



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Example: Kilpisjärvi summer temperature

Posterior of temperature change



Posterior sd ≈ 0.827 ; MCSE of the mean ≈ 0.0261

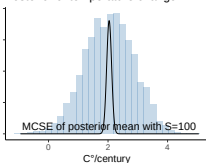
More draws reduce MCSE, but do not reduce posterior uncertainty.



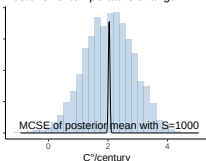
Example: Kilpisjärvi summer temperature

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Posterior of temperature change



Posterior of temperature change





Example: Kilpisjärvi summer temperature

- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

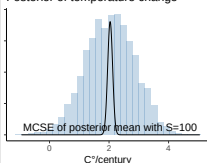
- Monte Carlo methods

- Direct sampling

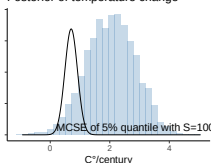
- Indirect sampling

- Rejection sampling
- Importance sampling
- Pareto-smoothed importance sampling

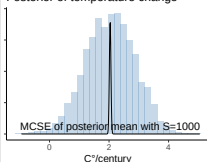
Posterior of temperature change



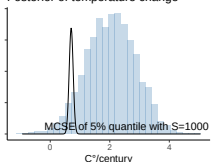
Posterior of temperature change



Posterior of temperature change



Posterior of temperature change





Example: Kilpisjärvi summer temperature

- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

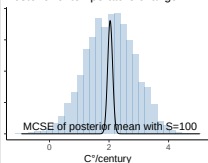
- Monte Carlo methods

- Direct sampling

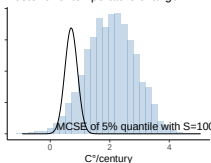
- Indirect sampling

- Rejection sampling
- Importance sampling
- Pareto-smoothed importance sampling

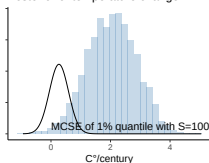
Posterior of temperature change



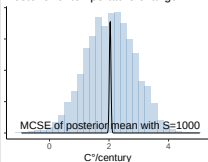
Posterior of temperature change



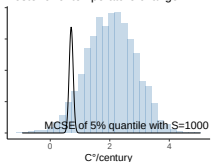
Posterior of temperature change



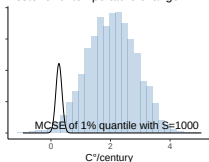
Posterior of temperature change



Posterior of temperature change



Posterior of temperature change





Example: Kilpisjärvi summer temperature

- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

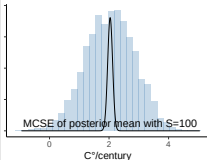
- Monte Carlo methods

- Direct sampling

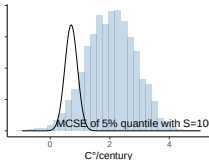
- Indirect sampling

- Rejection sampling
- Importance sampling
- Pareto-smoothed importance sampling

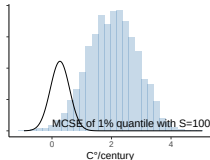
Posterior of temperature change



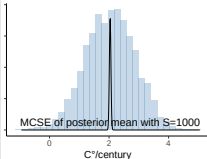
Posterior of temperature change



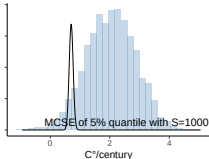
Posterior of temperature change



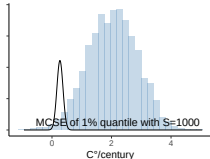
Posterior of temperature change



Posterior of temperature change



Posterior of temperature change



Tail quantiles are more difficult to estimate



Estimating a posterior probability

- Let $p_A = P(\theta \in A \mid y)$ and define $I_s = I(\theta^{(s)} \in A)$. Then

$$I_s \sim \text{Bernoulli}(p_A), \quad \hat{p}_A = \frac{1}{S} \sum_{s=1}^S I_s.$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Estimating a posterior probability

- Let $p_A = P(\theta \in A \mid y)$ and define $I_s = I(\theta^{(s)} \in A)$. Then

$$I_s \sim \text{Bernoulli}(p_A), \quad \hat{p}_A = \frac{1}{S} \sum_{s=1}^S I_s.$$

- The number of draws in A is $\sum_s I_s \sim \text{Binomial}(S, p_A)$, so

$$\text{MCSE}(\hat{p}_A) = \sqrt{\frac{p_A(1-p_A)}{S}} \approx \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{S}}.$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Estimating a posterior probability

- Let $p_A = P(\theta \in A \mid y)$ and define $I_s = I(\theta^{(s)} \in A)$. Then

$$I_s \sim \text{Bernoulli}(p_A), \quad \hat{p}_A = \frac{1}{S} \sum_{s=1}^S I_s.$$

- The number of draws in A is $\sum_s I_s \sim \text{Binomial}(S, p_A)$, so

$$\text{MCSE}(\hat{p}_A) = \sqrt{\frac{p_A(1-p_A)}{S}} \approx \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{S}}.$$

- If $S = 100$ and $p_A \approx 0.5$, MCSE is 0.05, or about 5 percentage points. An MCSE of 0.01 requires about 2500 draws.



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Estimating a posterior probability

- Let $p_A = P(\theta \in A \mid y)$ and define $I_s = I(\theta^{(s)} \in A)$. Then

$$I_s \sim \text{Bernoulli}(p_A), \quad \hat{p}_A = \frac{1}{S} \sum_{s=1}^S I_s.$$

- The number of draws in A is $\sum_s I_s \sim \text{Binomial}(S, p_A)$, so

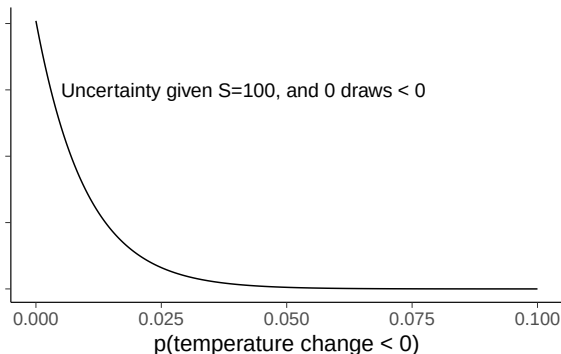
$$\text{MCSE}(\hat{p}_A) = \sqrt{\frac{p_A(1-p_A)}{S}} \approx \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{S}}.$$

- If $S = 100$ and $p_A \approx 0.5$, MCSE is 0.05, or about 5 percentage points. An MCSE of 0.01 requires about 2500 draws.
- Small probabilities require many draws
 - the expected number of draws in A is Sp_A , so we need S much larger than $1/p_A$ for a stable estimate



Example: Kilpisjärvi summer temperature

Posterior uncertainty $p(\text{temperature change} < 0)$

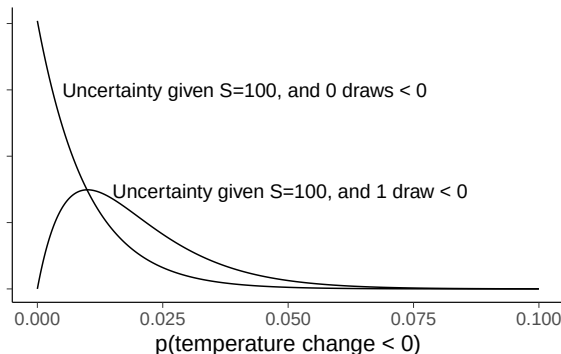


- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



Example: Kilpisjärvi summer temperature

Posterior uncertainty $p(\text{temperature change} < 0)$



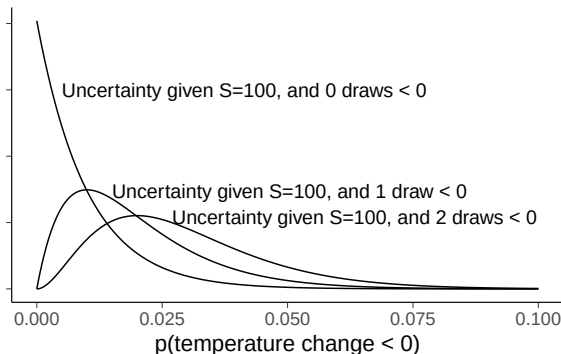
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



Example: Kilpisjärvi summer temperature

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

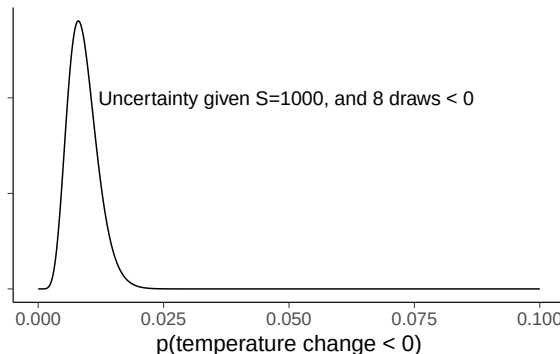
Posterior uncertainty $p(\text{temperature change} < 0)$





Example: Kilpisjärvi summer temperature

Posterior uncertainty $p(\text{temperature change} < 0)$



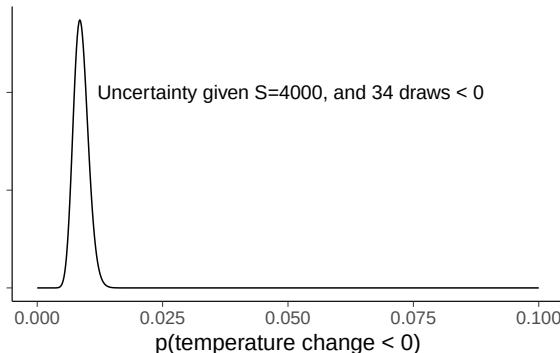
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



Example: Kilpisjärvi summer temperature

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Posterior uncertainty $p(\text{temperature change} < 0)$





UPPSALA
UNIVERSITET

How many digits to show in reports?

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Too many digits make reading of the results slower and give false impression of the accuracy



How many digits to show in reports?

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Too many digits make reading of the results slower and give false impression of the accuracy
 - Do not show digits that are just random noise
 - check what is the Monte Carlo standard error



How many digits to show in reports?

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Too many digits make reading of the results slower and give false impression of the accuracy
 - Do not show digits that are just random noise
 - check what is the Monte Carlo standard error
 - Show meaningful digits given the posterior uncertainty



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Too many digits make reading of the results slower and give false impression of the accuracy
- Do not show digits that are just random noise
 - check what is the Monte Carlo standard error
- Show meaningful digits given the posterior uncertainty
- Example: The mean and 90% central posterior interval for the temperature increase in $^{\circ}\text{C}/\text{century}$, based on posterior draws



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Too many digits make reading of the results slower and give false impression of the accuracy
- Do not show digits that are just random noise
 - check what is the Monte Carlo standard error
- Show meaningful digits given the posterior uncertainty
- Example: The mean and 90% central posterior interval for the temperature increase in $^{\circ}\text{C}/\text{century}$, based on posterior draws
 - 2.050774 and [0.7472868 3.3017524] (NO!)



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Too many digits make reading of the results slower and give false impression of the accuracy
- Do not show digits that are just random noise
 - check what is the Monte Carlo standard error
- Show meaningful digits given the posterior uncertainty
- Example: The mean and 90% central posterior interval for the temperature increase in $^{\circ}\text{C}/\text{century}$, based on posterior draws
 - 2.050774 and [0.7472868 3.3017524] (NO!)
 - 2.1 and [0.7 3.3]



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Too many digits make reading of the results slower and give false impression of the accuracy
- Do not show digits that are just random noise
 - check what is the Monte Carlo standard error
- Show meaningful digits given the posterior uncertainty
- Example: The mean and 90% central posterior interval for the temperature increase in $^{\circ}\text{C}/\text{century}$, based on posterior draws
 - 2.050774 and [0.7472868 3.3017524] (NO!)
 - 2.1 and [0.7 3.3]
 - 2 and [1 3] (depends on the context)



UPPSALA
UNIVERSITET

How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Example: The probability that the temperature increase is positive



UPPSALA
UNIVERSITET

How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Example: The probability that the temperature increase is positive
 - 0.9960000 (NO!)



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Example: The probability that the temperature increase is positive
 - 0.9960000 (NO!)
 - 1.00 (depends on the context)



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Example: The probability that the temperature increase is positive
 - 0.9960000 (NO!)
 - 1.00 (depends on the context)
 - With 4000 draws $MCSE \approx 0.002$. We could report that the probability is **very likely larger than 0.99**, or sample more to justify reporting three digits



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Example: The probability that the temperature increase is positive
 - 0.9960000 (NO!)
 - 1.00 (depends on the context)
 - With 4000 draws $MCSE \approx 0.002$. We could report that the probability is **very likely larger than 0.99**, or sample more to justify reporting three digits
 - For probabilities close to 0 or 1, also consider whether the model assumptions justify such precision



How many digits to show in reports?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Example: The probability that the temperature increase is positive
 - 0.9960000 (NO!)
 - 1.00 (depends on the context)
 - With 4000 draws $MCSE \approx 0.002$. We could report that the probability is **very likely larger than 0.99**, or sample more to justify reporting three digits
 - For probabilities close to 0 or 1, also consider whether the model assumptions justify such precision
- **For your project:** For each reported value, choose a sensible number of digits.



How many simulation draws are needed?

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Fewer draws may be needed with
 - deterministic methods
 - Rao-Blackwellization (marginalisation)
 - variance-reduction methods, such as control variates



How many simulation draws are needed?

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- The $S^{-1/2}$ Monte Carlo rate does not explicitly depend on dimension



How many simulation draws are needed?

- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

- Monte Carlo methods

- Direct sampling

- Indirect sampling

- Rejection sampling
- Importance sampling
- Pareto-smoothed
importance sampling

- The $S^{-1/2}$ Monte Carlo rate does not explicitly depend on dimension
- However, obtaining useful independent draws can be difficult in high-dimensional problems



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- **Direct sampling**
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Section 4

Direct sampling



- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - **Direct sampling**
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Direct simulation from a known density or mass function, such as $p(\theta | y)$ in a conjugate model
 - Produces independent draws
 - analytic transformations of uniform random numbers (e.g. Appendix A)
 - factorization
 - numerical inversion of the CDF
 - **Limitation:** available only for a limited set of models



UPPSALA
UNIVERSITET

Random number generators

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
 - **Direct sampling**
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- How do we sample from a probability distribution?



- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - **Direct sampling**
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- How do we sample from a probability distribution?
 - Good **pseudorandom** number generators are sufficient for Bayesian inference
 - a pseudorandom generator uses a deterministic algorithm to produce a sequence that behaves like a random sequence
 - modern statistical software uses high-quality generators



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- **Direct sampling**
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- **Box–Muller method:**

If U_1 and U_2 are independent draws from $\mathcal{U}(0, 1)$, and

$$X_1 = \sqrt{-2 \log(U_1)} \cos(2\pi U_2)$$

$$X_2 = \sqrt{-2 \log(U_1)} \sin(2\pi U_2)$$

then X_1 and X_2 are independent draws from the distribution $\mathcal{N}(0, 1)$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- **Direct sampling**
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Box–Muller method:

If U_1 and U_2 are independent draws from $\mathcal{U}(0, 1)$, and

$$X_1 = \sqrt{-2 \log(U_1)} \cos(2\pi U_2)$$

$$X_2 = \sqrt{-2 \log(U_1)} \sin(2\pi U_2)$$

then X_1 and X_2 are independent draws from the distribution $\mathcal{N}(0, 1)$

- it is not the fastest method because it requires trigonometric functions
- many alternative methods exist for the normal distribution
- R uses numerical inversion by default



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- **Indirect sampling**
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Section 5

Indirect sampling



UPPSALA
UNIVERSITET

Indirect sampling

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Rejection sampling



UPPSALA
UNIVERSITET

Indirect sampling

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Rejection sampling
- Importance sampling



UPPSALA
UNIVERSITET

Indirect sampling

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - **Indirect sampling**
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Rejection sampling
 - Importance sampling
 - Markov chain Monte Carlo (next week)



- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- The effective sample size S_{eff} compares Monte Carlo precision with that of independent, equally weighted draws
 - For S independent draws, $S_{\text{eff}} = S$



- Introduction

- Computational aspects

- Bayesian Computation

- Numerical integration

- Monte Carlo methods

- Direct sampling

- Indirect sampling

- Rejection sampling
- Importance sampling
- Pareto-smoothed importance sampling

- The effective sample size S_{eff} compares Monte Carlo precision with that of independent, equally weighted draws
- For S independent draws, $S_{\text{eff}} = S$
- Weighted or correlated draws usually have $S_{\text{eff}} < S$
- A larger S_{eff} means more precise estimates for the same number of stored draws



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Subsection 1

Rejection sampling



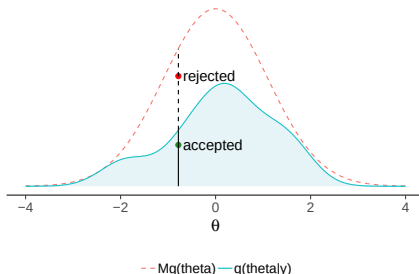
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Rejection sampling

- Let $p(\theta) = q(\theta)/Z$ be the target density
- Choose a proposal g and a finite M such that

$$q(\theta) \leq Mg(\theta) \quad \text{for all } \theta$$

- Intuitively, scale g vertically by M until Mg covers q everywhere. Then $q(\theta)/(Mg(\theta))$ is always between 0 and 1.





- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

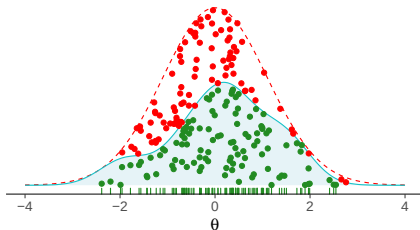
Rejection sampling

- Let $p(\theta) = q(\theta)/Z$ be the target density
- Choose a proposal g and a finite M such that

$$q(\theta) \leq Mg(\theta) \quad \text{for all } \theta$$

- Intuitively, scale g vertically by M until Mg covers q everywhere. Then $q(\theta)/(Mg(\theta))$ is always between 0 and 1.
- Draw $\theta^* \sim g$ and $u \sim \mathcal{U}(0, 1)$; accept if

$$u \leq \frac{q(\theta^*)}{Mg(\theta^*)}$$



• Accepted • Rejected - - Mg(theta) — q(theta)



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

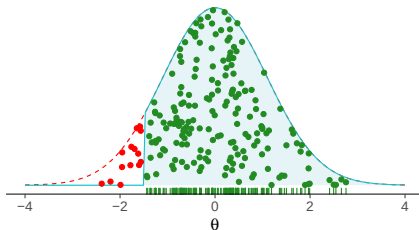
Rejection sampling

- Let $p(\theta) = q(\theta)/Z$ be the target density
- Choose a proposal g and a finite M such that

$$q(\theta) \leq Mg(\theta) \quad \text{for all } \theta$$

- Intuitively, scale g vertically by M until Mg covers q everywhere. Then $q(\theta)/(Mg(\theta))$ is always between 0 and 1.
- Draw $\theta^* \sim g$ and $u \sim \mathcal{U}(0, 1)$; accept if

$$u \leq \frac{q(\theta^*)}{Mg(\theta^*)}$$



• Accepted • Rejected - - Mg(theta) — q(theta)



Why rejection sampling works

- The density of an accepted proposal is proportional to

$$\underbrace{g(\theta)}_{\text{proposal density}} \underbrace{\frac{q(\theta)}{Mg(\theta)}}_{\text{acceptance probability}} = \frac{q(\theta)}{M} \propto p(\theta).$$

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



Why rejection sampling works

- The density of an accepted proposal is proportional to

$$\underbrace{g(\theta)}_{\text{proposal density}} \underbrace{\frac{q(\theta)}{Mg(\theta)}}_{\text{acceptance probability}} = \frac{q(\theta)}{M} \propto p(\theta).$$

- Thus the accepted values are independent draws from the target

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Why rejection sampling works

- The density of an accepted proposal is proportional to

$$\underbrace{g(\theta)}_{\text{proposal density}} \underbrace{\frac{q(\theta)}{Mg(\theta)}}_{\text{acceptance probability}} = \frac{q(\theta)}{M} \propto p(\theta).$$

- Thus the accepted values are independent draws from the target
- If A of T proposals are accepted, their effective sample size is A , while A/T estimates the acceptance rate Z/M
When will the acceptance rate be high or low?



Why rejection sampling works

- The density of an accepted proposal is proportional to

$$\underbrace{g(\theta)}_{\text{proposal density}} \underbrace{\frac{q(\theta)}{Mg(\theta)}}_{\text{acceptance probability}} = \frac{q(\theta)}{M} \propto p(\theta).$$

- Thus the accepted values are independent draws from the target
- If A of T proposals are accepted, their effective sample size is A , while A/T estimates the acceptance rate Z/M
 - A tight envelope gives a high acceptance rate
 - Finding a good envelope becomes difficult as dimension increases

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



UPPSALA
UNIVERSITET

Pair discussion: rejection proposal

- What happens if Mg fails to cover the target everywhere?
- What happens if the envelope is valid but much wider than the target?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Pair discussion: rejection proposal

- What happens if Mg fails to cover the target everywhere?
- What happens if the envelope is valid but much wider than the target?

After discussion:

- Without full coverage, the algorithm is invalid and misses part of the target.
- An overly wide envelope is valid but gives a low acceptance rate.
- A good envelope covers the target as tightly as possible.



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Subsection 2

Importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Importance sampling identity

If we can evaluate the normalized target p , then

$$w(\theta) = \frac{p(\theta)}{g(\theta)},$$

$$\mathbb{E}_p[f(\theta)] = \int f(\theta)w(\theta)g(\theta) d\theta = \mathbb{E}_g[w(\theta)f(\theta)].$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Importance sampling identity

If we can evaluate the normalized target p , then

$$w(\theta) = \frac{p(\theta)}{g(\theta)},$$

$$\mathbb{E}_p[f(\theta)] = \int f(\theta)w(\theta)g(\theta) d\theta = \mathbb{E}_g[w(\theta)f(\theta)].$$

Thus, for $\theta^{(s)} \sim g$,

$$\hat{l}_S = \frac{1}{S} \sum_{s=1}^S w(\theta^{(s)})f(\theta^{(s)})$$

is an unbiased estimator of $\mathbb{E}_p[f(\theta)]$.



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - **Importance sampling**
 - Pareto-smoothed importance sampling

Importance sampling identity

If we can evaluate the normalized target p , then

$$w(\theta) = \frac{p(\theta)}{g(\theta)},$$

$$\mathbb{E}_p[f(\theta)] = \int f(\theta)w(\theta)g(\theta) d\theta = \mathbb{E}_g[w(\theta)f(\theta)].$$

Thus, for $\theta^{(s)} \sim g$,

$$\hat{l}_S = \frac{1}{S} \sum_{s=1}^S w(\theta^{(s)})f(\theta^{(s)})$$

is an unbiased estimator of $\mathbb{E}_p[f(\theta)]$. If only $q(\theta) = Zp(\theta)$ is known, the unknown Z cancels:

$$\mathbb{E}_p[f(\theta)] = \frac{\mathbb{E}_g[(q/g)f]}{\mathbb{E}_g[q/g]}.$$

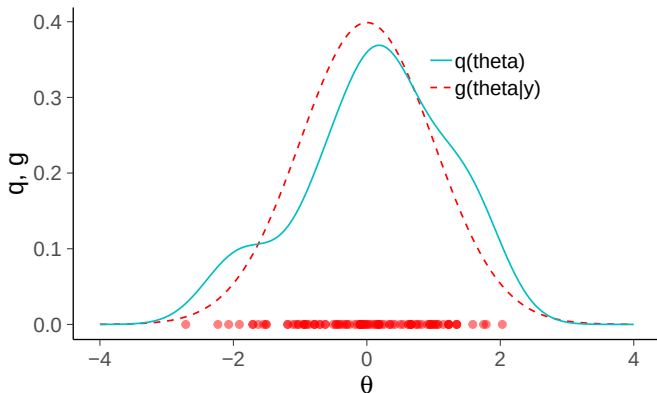


- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Self-normalized importance sampling

- Draw from g and compute $w_s = q(\theta^{(s)})/g(\theta^{(s)})$

Target, proposal, and draws



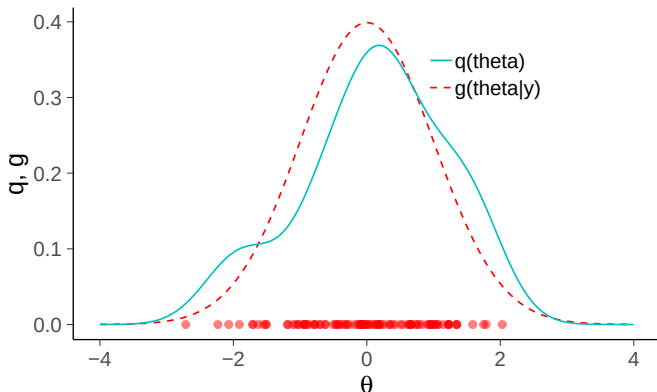


- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Self-normalized importance sampling

- Draw from g and compute $w_s = q(\theta^{(s)})/g(\theta^{(s)})$

Target, proposal, and draws



$$\widehat{E_p[f(\theta)]} = \sum_{s=1}^S \tilde{w}_s f(\theta^{(s)}), \quad \tilde{w}_s = \frac{w_s}{\sum_{r=1}^S w_r}$$

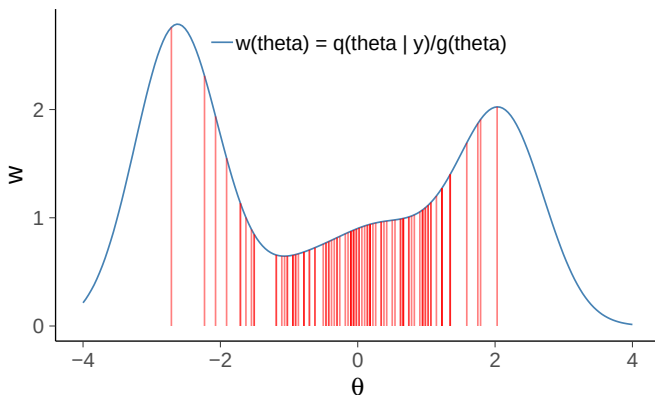


- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Self-normalized importance sampling

- Draw from g and compute $w_s = q(\theta^{(s)})/g(\theta^{(s)})$

Draws and importance weights



$$\widehat{E_p[f(\theta)]} = \sum_{s=1}^S \tilde{w}_s f(\theta^{(s)}), \quad \tilde{w}_s = \frac{w_s}{\sum_{r=1}^S w_r}$$

The estimator is generally biased for finite S , but consistent as $S \rightarrow \infty$.



UPPSALA
UNIVERSITET

Pair discussion: proposal tails

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- The target has heavier tails than the proposal.
- What happens to the importance weights in the tails?
- Would a slightly wider proposal be safer or less safe?



Pair discussion: proposal tails

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- The target has heavier tails than the proposal.
- What happens to the importance weights in the tails?
- Would a slightly wider proposal be safer or less safe?

After discussion:

- Rare tail draws receive extremely large weights.
- A few draws may dominate, producing low ESS and unstable estimates.
- A proposal with sufficiently heavy tails is usually safer.



Choosing a proposal

- The proposal must satisfy

$$g(\theta) > 0 \quad \text{wherever } p(\theta) > 0$$

Otherwise, important parts of the target can never be sampled.

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



Choosing a proposal

- The proposal must satisfy

$$g(\theta) > 0 \quad \text{wherever } p(\theta) > 0$$

Otherwise, important parts of the target can never be sampled.

- A good proposal covers the target and gives similar weights
 - tails that are too light can produce rare, extremely large weights
 - large weight variation gives unstable estimates and low effective sample size

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



Choosing a proposal

- The proposal must satisfy

$$g(\theta) > 0 \quad \text{wherever } p(\theta) > 0$$

Otherwise, important parts of the target can never be sampled.

- A good proposal covers the target and gives similar weights
 - tails that are too light can produce rare, extremely large weights
 - large weight variation gives unstable estimates and low effective sample size
- Finding a good proposal becomes harder as dimension increases

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Variation in the weights affects the **effective sample size**
 - if one weight dominates, we have effectively one draw
 - if all weights are equal, we have effectively S draws

Which weight pattern would you trust more?



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Variation in the weights affects the **effective sample size**
 - if one weight dominates, we have effectively one draw
 - if all weights are equal, we have effectively S draws
- For the usual central limit theorem, the weighted quantity must have finite variance; for self-normalized importance sampling this requires

$$E_g \left[w(\theta)^2 (f(\theta) - E_p[f(\theta)])^2 \right] < \infty.$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Variation in the weights affects the **effective sample size**
 - if one weight dominates, we have effectively one draw
 - if all weights are equal, we have effectively S draws
- For the usual central limit theorem, the weighted quantity must have finite variance; for self-normalized importance sampling this requires

$$E_g \left[w(\theta)^2 (f(\theta) - E_p[f(\theta)])^2 \right] < \infty.$$

- Heavy-tailed weights can violate this condition even when $f(\theta)$ itself has finite variance under the target.



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

1. Draw $(\alpha^{(s)}, \beta^{(s)})$ from the bivariate normal proposal g
2. Compute log weights
 $\log w_s = \log q(\alpha^{(s)}, \beta^{(s)}) - \log g(\alpha^{(s)}, \beta^{(s)})$
3. Stabilize and normalize the weights so that $\sum_s \tilde{w}_s = 1$
4. Compute weighted posterior summaries and inspect the weights and S_{eff}
5. Optionally resample draws with probabilities $\tilde{w}_s$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

1. Draw $(\alpha^{(s)}, \beta^{(s)})$ from the bivariate normal proposal g
2. Compute log weights
 $\log w_s = \log q(\alpha^{(s)}, \beta^{(s)}) - \log g(\alpha^{(s)}, \beta^{(s)})$
3. Stabilize and normalize the weights so that $\sum_s \tilde{w}_s = 1$
4. Compute weighted posterior summaries and inspect the weights and S_{eff}
5. Optionally resample draws with probabilities $\tilde{w}_s$

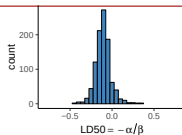
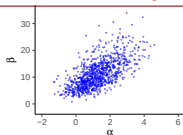
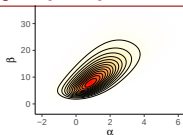
IR in the following figures means **importance resampling**:
resampling the proposal draws using the normalized weights.



UPPSALA
UNIVERSITET

Bioassay: proposal and resampled draws

Grid



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

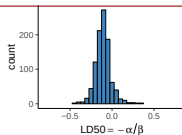
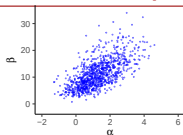
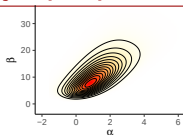


UPPSALA
UNIVERSITET

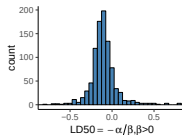
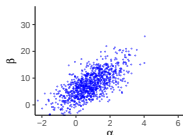
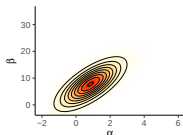
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Bioassay: proposal and resampled draws

Grid



Normal



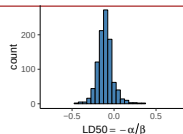
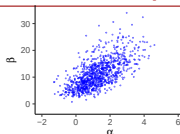
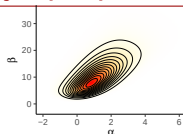
Normal approximation is discussed more in BDA3 Ch 4



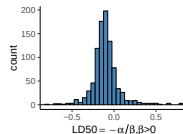
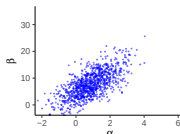
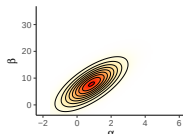
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Bioassay: proposal and resampled draws

Grid



Normal



Normal approximation is discussed more in BDA3 Ch 4
But the normal approximation is not accurate here:
 $Grid\ SD(LD50) \approx 0.1$, $normal\ SD(LD50) \approx 0.75$

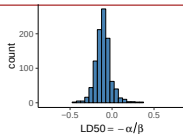
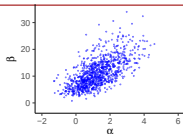
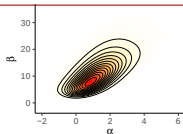


UPPSALA
UNIVERSITET

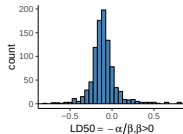
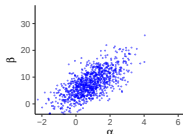
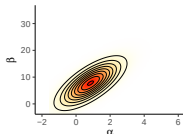
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Bioassay: proposal and resampled draws

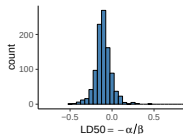
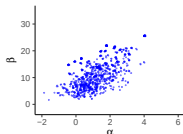
Grid



Normal



IR

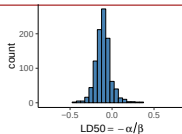
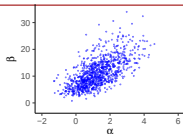
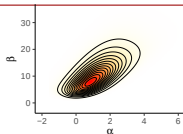




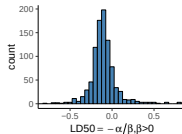
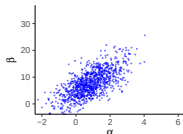
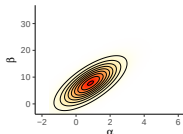
- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Bioassay: proposal and resampled draws

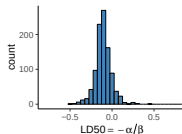
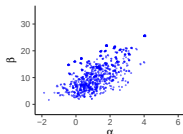
Grid



Normal



IR

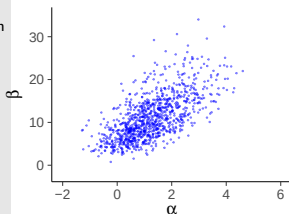


Grid $SD(LD50) \approx 0.1$, IR $SD(LD50) \approx 0.1$

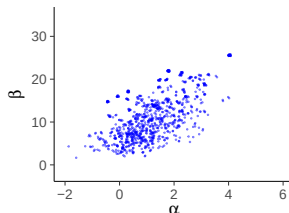


- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Grid



IR (resampled)



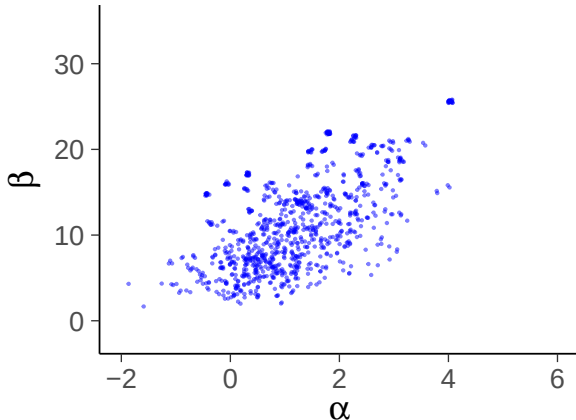


UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

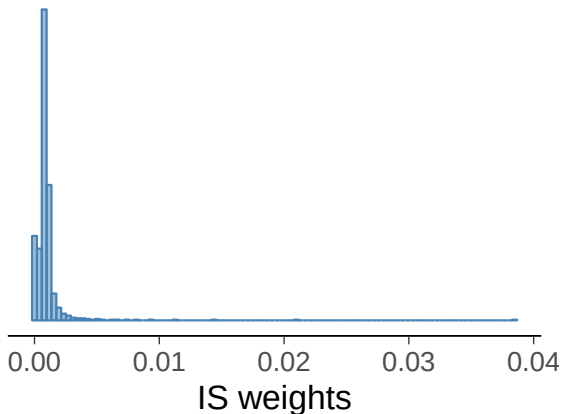
Bioassay: importance-resampled draws

IR (resampled)



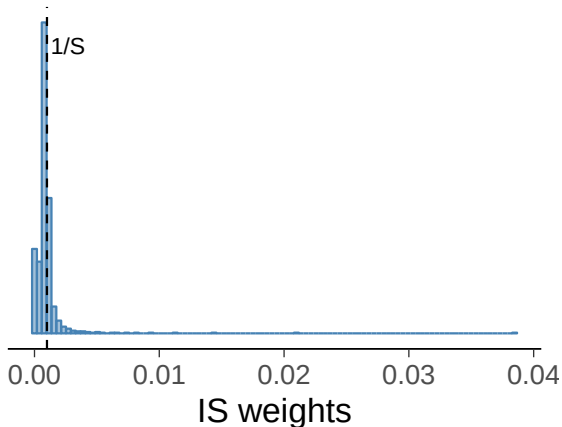


- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling





- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling





Pair discussion: can we trust the weights?

- Most normalized weights are near zero, while a few are much larger.
- What does this imply about the estimate?
- What should we inspect before trusting it?

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Pair discussion: can we trust the weights?

- Most normalized weights are near zero, while a few are much larger.
- What does this imply about the estimate?
- What should we inspect before trusting it?

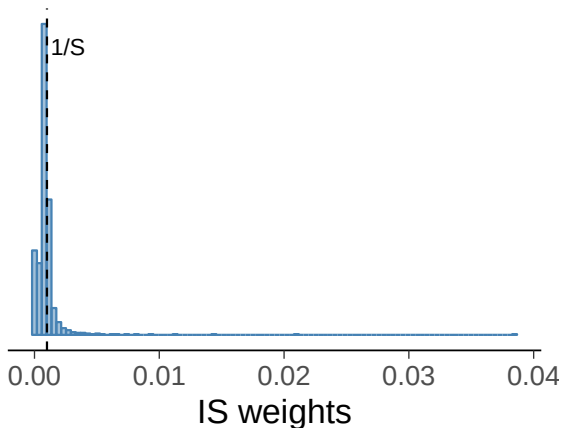
After discussion:

- Only a small number of draws contribute substantial information.
- The estimate may be sensitive to a few draws.
- Inspect ESS and the Pareto $\hat{k}$ diagnostic; improve the proposal if necessary.



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Bioassay: effective sample size

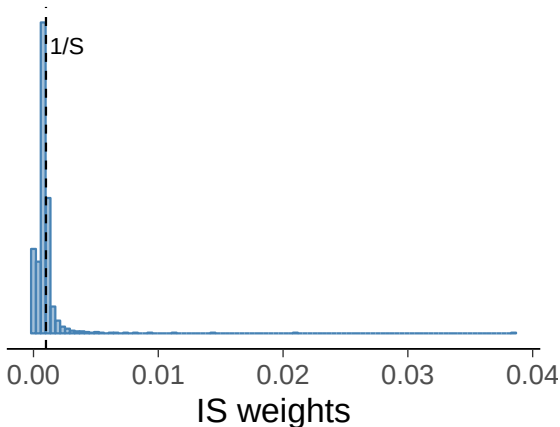


$$S_{\text{eff}} = \frac{1}{\sum_{s=1}^S (\tilde{w}(\theta^s))^2}, \quad \text{where } \tilde{w}(\theta^s) = w(\theta^s) / \sum_{s'=1}^S w(\theta^{s'})$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Bioassay: effective sample size



$$S_{\text{eff}} = \frac{1}{\sum_{s=1}^S (\tilde{w}(\theta^s))^2}, \quad \text{where } \tilde{w}(\theta^s) = w(\theta^s) / \sum_{s'=1}^S w(\theta^{s'})$$

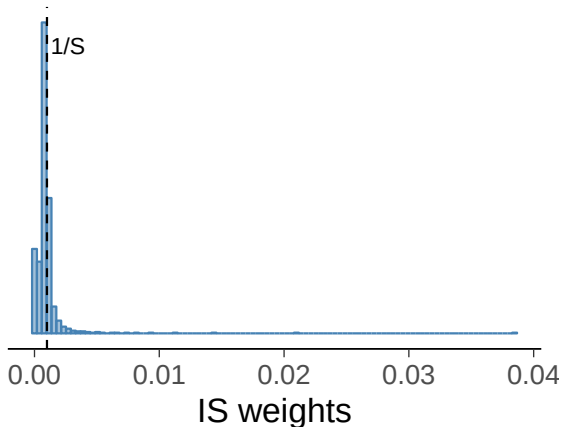
BDA3 1st (2013) and 2nd (2014) printing have an error for $\tilde{w}(\theta^s)$. The normalized weights equation should not have the multiplier S (the normalized weights should sum to one). Errata for the book

http://www.stat.columbia.edu/~gelman/book/errata_bda3.txt



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Bioassay: effective sample size



$$S_{\text{eff}} = \frac{1}{\sum_{s=1}^S (\tilde{w}(\theta^s))^2}, \quad \text{where } \tilde{w}(\theta^s) = w(\theta^s) / \sum_{s'=1}^S w(\theta^{s'})$$
$$S_{\text{eff}} \approx 270$$



UPPSALA
UNIVERSITET

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Subsection 3

Pareto-smoothed importance sampling



UPPSALA
UNIVERSITET

Pareto-smoothed importance sampling

- Raw importance sampling can be dominated by a few very large weights

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Raw importance sampling can be dominated by a few very large weights
- PSIS fits a generalized Pareto distribution to the upper tail of the importance ratios
- It replaces the largest ratios with smoothed values from the fitted tail



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Pareto-smoothed importance sampling

- Raw importance sampling can be dominated by a few very large weights
- PSIS fits a generalized Pareto distribution to the upper tail of the importance ratios
- It replaces the largest ratios with smoothed values from the fitted tail
- This usually reduces variability, at the cost of a small amount of bias
- PSIS also provides improved estimates of effective sample size and MCSE

Vehtari, Simpson, Gelman, Yao, and Gabry (2024)



- $\hat{k}$ describes the heaviness of the upper tail of the importance ratios
 - smaller $\hat{k}$ means more stable importance sampling
 - larger $\hat{k}$ means that a few ratios dominate

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

The Pareto $\hat{k}$ diagnostic

- $\hat{k}$ describes the heaviness of the upper tail of the importance ratios
 - smaller $\hat{k}$ means more stable importance sampling
 - larger $\hat{k}$ means that a few ratios dominate
- Current guidance treats estimates as reliable when

$$\hat{k} < \min\left(1 - \frac{1}{\log_{10} S}, 0.7\right)$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

The Pareto $\hat{k}$ diagnostic

- $\hat{k}$ describes the heaviness of the upper tail of the importance ratios
 - smaller $\hat{k}$ means more stable importance sampling
 - larger $\hat{k}$ means that a few ratios dominate
- Current guidance treats estimates as reliable when

$$\hat{k} < \min\left(1 - \frac{1}{\log_{10} S}, 0.7\right)$$

- Above this threshold, the estimate and its MCSE may be unreliable; investigate the proposal or use a more robust computation



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

The Pareto $\hat{k}$ diagnostic

- $\hat{k}$ describes the heaviness of the upper tail of the importance ratios
 - smaller $\hat{k}$ means more stable importance sampling
 - larger $\hat{k}$ means that a few ratios dominate
- Current guidance treats estimates as reliable when

$$\hat{k} < \min\left(1 - \frac{1}{\log_{10} S}, 0.7\right)$$

- Above this threshold, the estimate and its MCSE may be unreliable; investigate the proposal or use a more robust computation
- For intuition: a generalized Pareto tail has finite variance when $k < 1/2$; when $k \geq 1$, even its mean does not exist



Importance-sampling leave-one-out cross-validation

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Later in the course, we use $p(\theta | y)$ as a proposal distribution for $p(\theta | y_{-i})$
 - this allows fast approximation of leave-one-out cross-validation

$$p(y_i | y_{-i}) = \int p(y_i | \theta) p(\theta | y_{-i}) d\theta$$



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Recap: simulation methods

Method	Draw from	Output	Main limitation
Direct	Target p	Independent, equal-weight draws	Available for few models
Rejection	Proposal g	Accepted independent target draws	Acceptance can be very low
Importance	Proposal g	Weighted proposal draws	A few weights may dominate
MCMC	Transition rule	Dependent target draws	Requires convergence diagnostics



- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

Recap: simulation methods

Method	Draw from	Output	Main limitation
Direct	Target p	Independent, equal-weight draws	Available for few models
Rejection	Proposal g	Accepted independent target draws	Acceptance can be very low
Importance	Proposal g	Weighted proposal draws	A few weights may dominate
MCMC	Transition rule	Dependent target draws	Requires convergence diagnostics

- There is no universally best method: efficiency depends on the target, dimension, proposal, and quantity being estimated.
- Always assess Monte Carlo accuracy with appropriate diagnostics.



Next week: Markov chain Monte Carlo (MCMC)

- Introduction
 - Computational aspects
 - Bayesian Computation
 - Numerical integration
 - Monte Carlo methods
 - Direct sampling
 - Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling
- Pros
 - Markov chain goes where most of the posterior mass is
 - Certain MCMC methods scale well to high dimensions
 - Cons
 - Draws are dependent (affects how many draws are needed)
 - Convergence in practical time is not guaranteed



Next week: Markov chain Monte Carlo (MCMC)

- Introduction
 - Computational aspects
- Bayesian Computation
 - Numerical integration
- Monte Carlo methods
- Direct sampling
- Indirect sampling
 - Rejection sampling
 - Importance sampling
 - Pareto-smoothed importance sampling

- Pros
 - Markov chain goes where most of the posterior mass is
 - Certain MCMC methods scale well to high dimensions
- Cons
 - Draws are dependent (affects how many draws are needed)
 - Convergence in practical time is not guaranteed
- MCMC methods in this course
 - Gibbs sampling: “iterative conditional sampling”
 - Metropolis: “random walk in joint distribution”
 - Dynamic Hamiltonian Monte Carlo: “state-of-the-art” used in Stan